

Muhammad Inshal

 mInshal852 |  m-inshal |  inshalrrr@gmail.com |  +92-3157447900

SUMMARY

Python Backend Engineer with a B.S. in Computer Science and hands-on experience building and deploying scalable backend services using FastAPI, PostgreSQL, MongoDB, and Docker. Skilled in REST API development, AI integration, and CI/CD automation via GitHub Actions on Linux environments.

EDUCATION

FAST-NUCES, Faisalabad

07/2022 – 06/2026

Bachelor of Science in Computer Science

Coursework: OOP, Data Structures & Algorithms, Databases, Operating Systems, Computer Networks, AI, GenAI

ACHIEVEMENTS

Solved 250+ Data Structures and Algorithms problems on LeetCode, demonstrating strong problem-solving and algorithmic thinking skills.

CERTIFICATIONS

Retrieval-Augmented Generation (RAG): Built end-to-end RAG applications covering dataset creation, web scraping (Zyte API), chunking, embeddings, ChromaDB vector indexing, semantic retrieval, prompt engineering, and LLM integration.

Docker: Containerization, Docker images, Docker Compose, volumes, networking, and application deployment.

Git & GitHub: Version control, branching, merging, pull requests, and collaborative software development workflows.

EXPERIENCE

Software Engineering Trainee — FAST-NUCES, Faisalabad

Summer 2025

- Trained under faculty supervision in software engineering practices including MVC architecture, MVP development, SDLC, and writing clean, maintainable code.
- Contributed to an open-source educational storytelling platform for Palestinian children, supporting development of an offline/online web app for interactive language learning via Git-based collaborative workflows.

PROJECTS

TestHub (FYP) — React, TypeScript, Go, MongoDB, Docker, FastAPI, Gemini 

- Developed a web-based automated testing platform supporting Selenium and Playwright, with parallel execution via Selenium Grid and Docker, and a centralized dashboard for results, logs, screenshots, and reports.
- Engineered an AI-powered failure diagnosis module using FastAPI and Google Gemini API to analyze logs, stack traces, and screenshots, providing root-cause analysis and debugging recommendations.
- Containerized all services with Docker and implemented a GitHub Actions CI/CD pipeline for automated builds, testing, and deployment with downloadable artifacts.

Country FACT RAG — FastAPI, Streamlit, ChromaDB, Sentence Transformers, OpenRouter, Docker 

- Built an end-to-end RAG system answering country-related questions over a custom knowledge base scraped from Britannica via the Zyte API.
- Built the ingestion pipeline: chunking, Sentence Transformer embeddings, and HNSW indexing in ChromaDB for semantic retrieval.
- Built a FastAPI backend with query rewriting and answer generation via OpenRouter (Qwen), served through a Streamlit UI and deployed with Docker Compose.

Neural Storyteller: Image Captioning — PyTorch, torchvision 

- Built an end-to-end image captioning pipeline using ResNet50 for feature extraction and an LSTM-based Seq2Seq decoder for natural language generation.
- Implemented greedy decoding and beam search strategies to enhance caption quality, with a Gradio web app for real-time inference.

Masked Autoencoder (MAE) — PyTorch, Vision Transformer

- Designed a self-supervised MAE with an asymmetric ViT architecture: ViT-Base (768-dim) encoding 25% visible patches and a lightweight ViT-Small (384-dim) decoder reconstructing masked 75%.
- Built a patchify/unpatchify and random masking pipeline for 224×224 images into 196 tokens of 16×16 patches; trained with Masked MSE Loss on TinyImageNet.

SKILLS

Languages Python, SQL, Go, C++, C#

Backend FastAPI, REST APIs, PostgreSQL, MongoDB, ChromaDB

DevOps Docker, CI/CD, Git, GitHub, GitHub Actions, Linux (Ubuntu)

AI/ML PyTorch, Retrieval-Augmented Generation (RAG), Sentence Transformers, ChromaDB, OpenRouter API, Google Gemini API

Tools Oracle Database, Figma, Kaggle